WHICH FINE-TUNING CHANGES MATTER?
PROBING REPRESENTATION SHIFTS IN ACCENTED SPEECH RECOGNITION

TODO: Author names

TODO: Affiliations and contact information

ABSTRACT

Fine-tuning changes speech representations, but recognition accuracy alone does not reveal which changes an adapted model uses. We study this dependence by selectively reverting final-layer coordinates to their values before adaptation while freezing the adapted CTC head. Decision-Aligned Delta Selection ranks these changes using the product of their displacement and CTC-loss gradient. On two L2-ARCTIC splits, reverting the lowest-ranked quarter of wav2vec 2.0 shift coordinates increases word error rate by 0.22 and 0.08 percentage points; reverting the highest-ranked quarter increases it by 7.10 and 9.86 points. A data2vec checkpoint exhibits the same deletion asymmetry, but rescaled random selection outperforms the Taylor ranking at 50% retention. These results establish uneven functional dependence on representation shifts in the evaluated models, while showing that a useful deletion ranking need not yield the best retained representation at every budget.

Index Terms— speech recognition, self-supervised learning, model adaptation, representation analysis, CTC

1. INTRODUCTION

Adapting a speech recognizer to non-native speakers requires it to accommodate pronunciation variability while preserving linguistic distinctions. Self-supervised speech encoders provide a strong starting point for this adaptation [1, 2]. Supervised contrastive regularization, for example, uses repeated transcripts to improve accented speech recognition [3]. Yet the resulting word error rate (WER) summarizes the adapted system as a whole. It does not tell us whether the recognizer relies on changes distributed throughout its representation or on a more selective set of changes. Understanding this dependence would help distinguish what fine-tuning changes from what the learned recognition rule actually uses.

Representation analyses offer part of this picture. Layer-wise probes and similarity measures characterize acoustic and linguistic information and its evolution during ASR fine-tuning [4, 5]. Multitask fine-tuning can also preserve acoustic and semantic information in separate final-layer branches [6]. Such studies describe the information available to downstream tasks. A complementary question concerns the changes used by a particular adapted recognizer, whose decision rule has already been learned.

Fine-tuning differences provide a natural object for this question. In parameter space, task arithmetic uses differences between source and adapted weights to edit model behavior [7]. At the representation level, *Mind the Shift* shows that differences between fine-tuned and pretrained speech embeddings can improve child ASR when fused with another encoder [8]. Its newly trained fusion head establishes that the complete difference contains useful information. It leaves open how an existing recognizer depends on individual components of its own representation change. Likewise, learnable layer and dimension-wise source selection [9] optimizes a downstream representation, allowing the readout to adjust to the selected inputs.

We instead examine the adapted recognizer directly. For the same waveform, we subtract the representation before target-domain adaptation from the adapted representation. We then retain selected coordinates of this shift and supply the resulting representation to the original, frozen CTC head. A removed shift coordinate reverts to its source value. No readout is refitted. To order the coordinates, we use a Taylor saliency criterion [10] that combines displacement with sensitivity to the CTC loss. We call this procedure *Decision-Aligned Delta Selection* (DADS). Its role is to probe dependence on the shift, with magnitude selection serving as a matched-budget control.

Our experiments support two findings. First, two locally adapted wav2vec 2.0 models retain nearly their full-shift WER after the lowest-ranked quarter of coordinates is reverted, while reverting the highest-ranked quarter causes a much larger loss. Second, a data2vec model shares this asymmetry but favors rescaled random masks over DADS at 50% retention. The contribution is thus an intervention-based account of uneven dependence on final-layer adaptation changes, together with evidence that coordinate importance and retained-representation quality can diverge. We study this dependence in each encoder's native coordinate basis on L2-ARCTIC, using checkpoint pairs with recorded initialization.

2. PROBING FINE-TUNING SHIFTS

2.1. Paired representations and coordinate reversion

Let f_0 be the encoder before target-domain adaptation and f_{ft} its adapted descendant. Given waveform x, their final-layer representations are

$$E_0(x), E_{\text{ft}}(x) \in \mathbb{R}^{T_x \times d}, \qquad \Delta(x) = E_{\text{ft}}(x) - E_0(x). \quad (1)$$

Both encoders receive the same input and preserve frame alignment. The source models in our experiments already include LibriSpeech ASR training; 0 denotes the state before L2-ARCTIC adaptation, rather than an exclusively self-supervised checkpoint. Unlike a parameter task vector, $\Delta(x)$ depends on the input waveform.

Let $g_{\text{ft}}(h) = W_{\text{ft}} h + b_{\text{ft}}$ be the adapted linear CTC head, applied frame by frame. For a binary mask $m \in \{0,1\}^d$, shared across frames and utterances, define

$$\widetilde{E}_m(x) = E_0(x) + m \odot \Delta(x), \qquad \hat{y}_m = \operatorname{Dec}\big(g_{\text{ft}}(\widetilde{E}_m(x))\big), \quad (2)$$

where Dec is greedy CTC decoding. A retained coordinate equals its adapted value; a reverted coordinate equals its source value. Thus $m = \mathbf{1}$ recovers the adapted representation and $m = \mathbf{0}$ supplies the source representation to the adapted head. We call these endpoints *Full* and *NoShift*. NoShift measures compatibility with the fixed adapted head; evaluating the original recognizer would require its original head as well.

All masks retain a d-dimensional representation and use the same encoders and readout. The intervention changes which adaptation displacements are supplied, without reducing encoder parameter count. This construction also requires a valid source–descendant pair: matching hidden dimensions and satisfying $E_0 + \Delta = E_{\text{ft}}$ alone cannot establish checkpoint lineage.

2.2. Decision-aligned ranking

We estimate coordinate importance on a labeled calibration subset $\mathcal{D}_u$ of the adaptation training set. Define the transcript-length-normalized loss

$$\ell_{x,y}(\Delta) = \frac{1}{|y|} \mathcal{L}_{\text{CTC}}\big(g_{\text{ft}}(E_0(x) + \Delta(x)), y\big), \quad (3)$$

where $|y|$ is the number of target characters. CTC sums over valid frame-to-label alignments [11]. Gradients are evaluated at the full adapted representation; neither model parameters nor the calibration examples are updated.

Reverting coordinate i at all frames changes its values by $-\Delta_{t,i}$. The first-order approximation to the loss change is

$$\delta \ell_i \approx -\sum_{t=1}^{T_x} \Delta_{t,i}(x) \frac{\partial \ell_{x,y}}{\partial \Delta_{t,i}(x)}. \quad (4)$$

We rank coordinates by the following nonnegative score:

$$U_i = \frac{1}{N_u} \sum_{(x,y)\in\mathcal{D}_u} \sum_{t=1}^{T_x} \left| \Delta_{t,i}(x) \frac{\partial \ell_{x,y}}{\partial \Delta_{t,i}(x)} \right|. \quad (5)$$

Here $N_u = \sum_{(x,y)\in\mathcal{D}_u} T_x$. The absolute value is taken before summing over frames, avoiding cancellation between positive and negative contributions. Consequently, U_i is an aggregate local saliency score, rather than the signed loss change in Eq. 4. Its usefulness for a finite coordinate reversion is assessed by recognition error. Padded frames are excluded; all valid frames, including those involved in blank decisions, contribute.

The magnitude and gradient controls use the same aggregation:

$$M_i = \frac{1}{N_u} \sum_{(x,y),t} |\Delta_{t,i}(x)|, \quad (6)$$

$$G_i = \frac{1}{N_u} \sum_{(x,y),t} |\partial \ell_{x,y} / \partial \Delta_{t,i}(x)|. \quad (7)$$

These controls isolate displacement size and loss sensitivity, respectively. DADS combines both, following the established use of activation–gradient products for saliency [10].

2.3. Retention and deletion

For retention, we keep the K highest-ranked coordinates and revert the remaining $d - K$. This tests how much recognition performance the selected changes preserve. For deletion, we start from Full and revert either the highest- or lowest-ranked Q coordinates. The difference between these two conditions tests whether the ranking distinguishes changes with unequal effects on the fixed readout. Removing the lowest-ranked quarter is exactly the same mask as retaining the highest-ranked 75%; these are two interpretations of one measurement.

Random masks retain K uniformly selected coordinates. A further control rescales their displacements by d/K:

$$\widetilde{E}_m^{\text{RS}} = E_0 + \frac{d}{K} m \odot \Delta. \quad (8)$$

For uniform fixed-size masks, this preserves the expected displacement over mask draws. It probes the effect of attenuating the total shift, although an individual rescaled coordinate can move beyond its adapted value.

3. EXPERIMENTAL SETUP

3.1. Data and adapted models

We use L2-ARCTIC, a corpus of non-native English read speech [12], with the released split CSVs of Thai et al. [3]. Each evaluated split has 18 training/development speakers

Table 1. Utterance counts in the evaluated released splits. Calibration is a subset of training. W1/W2 denote local wav2vec 2.0 models; D0 denotes local data2vec.

Model / fold	Train	Dev	Test	Calib.
W1 / 1	16,070	1,979	686	1,615
W2 / 2	15,953	2,086	691	1,600
D0 / 0	16,312	1,867	675	1,640

and six disjoint test speakers, spanning the same six first-language groups. Training and development share speakers. The normalized transcript strings do not overlap across train, development, and test. Some canonical prompt identifiers do overlap, so this is not a strictly unseen-prompt evaluation. Table 1 gives the actual manifest sizes.

W1 and W2 start from the LibriSpeech-960h wav2vec 2.0 Large checkpoint and use local CTC plus supervised contrastive fine-tuning. D0 starts from data2vec Audio Large, also trained on LibriSpeech-960h, and uses CTC-only adaptation. All three encoders have 24 Transformer layers and $d = 1024$, and inherit a 32-symbol character vocabulary and CTC head. We use the original head from each resulting checkpoint. These are separate adaptation settings, not a matched comparison of pretraining objectives.

Both recipes warm up the head for one epoch at learning rate 3×10^{-6}, followed by joint adaptation at 10^{-5} with the convolutional feature encoder frozen. The maximum joint budget is 40 epochs and selection uses development WER. W1/W2 use transcript-grouped batches of 24 per GPU on two GPUs, contrastive weight 0.05, temperature 0.1, projection dimension 256, and a linear learning-rate schedule. D0 uses batches of four on one GPU and a cosine schedule. The selected joint checkpoints occur at steps 2,204, 1,406, and 16,312 for W1, W2, and D0, respectively. Each is one adaptation run with seed 1337. W1/W2 training clips audio to 10 s and targets to 128 characters; shift evaluation uses complete utterances.

3.2. Calibration and evaluation

Calibration takes every tenth utterance after sorting by utterance identifier within each training speaker. This subset is part of encoder training, rather than held out from adaptation. Its labels determine a single ranking per checkpoint; development and test labels are not used to calculate the scores. We report the recorded retention budgets and compare them within each checkpoint.

Both encoder passes use the same 16-kHz waveform and the source processor, in evaluation mode. Features are cached in float32. Full-shift reconstruction agrees with the adapted features within 4.77×10^{-7} for W1/W2 and 1.20×10^{-7} for D0; maximum logit differences are below 7.63×10^{-6}. All conditions use greedy CTC decoding without a language model. WER is total word edit distance divided by total reference words, reported as a percentage. References use the project English normalizer; decoded hypotheses are lowercased and whitespace-tokenized. We report comparisons within this common evaluation path.

Table 2. Test WER (%) under the original frozen head. Retention percentages refer to shift coordinates; all inputs remain 1024-dimensional.

Condition	W1	W2	D0
NoShift	20.33	19.15	13.77
Full	15.14	14.46	11.39
DADS, 75%	15.36	14.54	10.96
DADS, 50%	16.01	15.58	12.94
Magnitude, 50%	16.63	16.48	13.75

W1/W2 have Full, NoShift, 75% and 50% DADS retention, 50% magnitude retention, and high/low-quarter deletion results. D0 additionally has 25% retention, gradient-only, random, and rescaled-random controls. Random results give mean and sample standard deviation over masks with seeds 1337, 2027, and 31415; these are not repeated training runs.

4. RESULTS AND ANALYSIS

4.1. How much of the shift preserves recognition?

Table 2 compares the endpoints and retained shifts. Keeping 75% of DADS-ranked coordinates gives WERs of 15.36% and 14.54% for W1 and W2, close in absolute terms to Full at 15.14% and 14.46%. Keeping half raises WER by 0.87 and 1.12 percentage points. The same budget selected by magnitude is worse by a further 0.62 and 0.91 points. Thus, loss sensitivity improves on displacement size alone for these retained masks, while full-shift performance is not completely recovered at 50%.

For D0, 75% retention yields 10.96% WER versus 11.39% for Full, whereas 50% gives 12.94%. These point estimates show that more retained change need not give lower error. The NoShift endpoints are worse than Full for all three models, establishing that the adapted head benefits from the complete representation change. We use the WER difference from Full to quantify the cost of reversion; the NoShift gap also reflects the head's compatibility with its inputs.

4.2. Which reverted changes are most damaging?

Figure 1 gives the complementary deletion comparison. Removing the highest-ranked quarter raises WER to 22.24%, 24.32%, and 14.64% for W1, W2, and D0. Removing the lowest-ranked quarter gives 15.36%, 14.54%, and 10.96%, respectively. The separation is much larger than the small departures from Full after low-ranked changes are reverted.

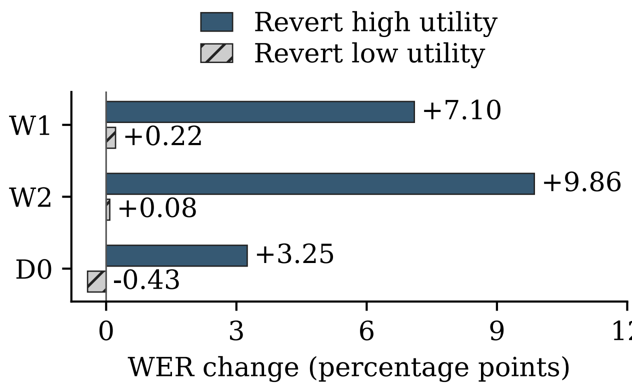

Fig. 1. Change in test WER after reverting 25% of shift coordinates, relative to Full. High/low refers to the DADS ranking. Negative values indicate an improvement. Each pair uses one fixed adapted recognizer.

Table 3. D0 test WER (%) at three retention budgets. Random entries show mean ± sample SD over three masks of the same checkpoint. Full: 11.39%.

Method	25%	50%	75%
DADS	15.20	12.94	10.96
Magnitude	13.30	13.75	11.10
Gradient	15.41	13.68	11.11
Random	13.00 ± 0.03	15.08 ± 0.31	11.35 ± 0.08
Random scaled	12.58 ± 0.37	11.95 ± 0.33	11.58 ± 0.07

This is direct evidence of uneven dependence on groups of shift coordinates under the fixed decision rule. It goes beyond observing unequal shift magnitudes or probe scores. It does not establish that each high-ranked coordinate is individually necessary, because coordinates are removed jointly and their effects can interact. In addition, the 75% retained representation still uses source values in all reverted coordinates; the retained shift is not a stand-alone representation of speech.

4.3. Does the Taylor ranking always retain the best shift?

The wider D0 comparison in Table 3 separates the usefulness of a deletion ranking from its quality at a particular retention budget. DADS has the lowest observed WER at 75%, but at 50% the rescaled random control reaches 11.95 ± 0.33%, compared with 12.94% for DADS. At 25%, both magnitude and random selection also outperform DADS. Its ranking therefore does not define a consistently superior sequence of retained representations.

Rescaling reduces the mean 50% random-mask WER from 15.08% to 11.95%. Thus, changing displacement scale substantially improves these fixed random masks, making unscaled random selection an incomplete control for the ranking. The comparison does not isolate a single adaptation mechanism: coordinate composition and displacement scale both influence the fixed head, and the Taylor score is evaluated locally at Full.

Existing D0 calibration-size results give a further check. With 128 calibration utterances, 75% DADS retention yields a mean 11.11% WER across three sampled subsets, compared with 10.96% using all 1,640 utterances. At 50%, the corresponding values are 13.11% and 12.94%. The 50% deficit relative to rescaled random masks remains even with the complete calibration set, so it cannot be attributed simply to using a very small subset.

5. DISCUSSION AND CONCLUSION

The common finding across these three checkpoints is selective dependence on final-layer changes: reverting low-ranked groups has a much smaller effect than reverting equally sized high-ranked groups. DADS supplies a simple way to expose this dependence. Its advantage over magnitude at 50% on W1/W2 is a useful selection result, while the D0 controls demonstrate that deletion asymmetry is a more consistent finding than a single ranking's superiority across budgets.

These interventions characterize the adapted CTC decision rule in the learned coordinate basis. They do not separate phonetic adaptation from readout compatibility, identify independent linguistic features, or establish the behavior of other layers. W1/W2 share a source model and training recipe, whereas D0 changes both the encoder family and adaptation objective. Broader architectural conclusions require those factors to be varied separately. The current comparisons also describe single training runs on one read-speech corpus; they do not measure training variability or transfer to child speech.
TODO before submission: add paired uncertainty estimates for the reported local checkpoints and complete the W1/W2 random and rescaling controls.

Fine-tuning shifts therefore provide a useful object for studying recognition behavior when their checkpoint lineage and readout are fixed. In the evaluated models, a substantial group of changes can be reverted with little change in WER, but effective selection depends on the model and retention budget. Following up this distinction with matched scale controls and independent adaptation runs would clarify which aspects of the observed dependence persist beyond the present checkpoints.

6. REFERENCES

[1] Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Advances in Neural Information Processing Systems*, 2020, vol. 33, pp. 12449–12460.

[2] Alexei Baevski, Wei-Ning Hsu, Qiantong Xu, Arun Babu, Jiatao Gu, and Michael Auli, "data2vec: A general framework for self-supervised learning in speech, vision and language," in *Proceedings of the 39th International Conference on Machine Learning*. 2022, vol. 162 of *Proceedings of Machine Learning Research*, pp. 1298–1312, PMLR.

[3] Van-Phat Thai, Aradhya Dhruv, Duc-Thinh Pham, and Sameer Alam, "Contrastive regularization for accent-robust ASR," arXiv:2605.03297, 2026, Accepted at Interspeech 2026.

[4] Ankita Pasad, Ju-Chieh Chou, and Karen Livescu, "Layer-wise analysis of a self-supervised speech representation model," in *2021 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, 2021, pp. 914–921.

[5] Ankita Pasad, Bowen Shi, and Karen Livescu, "Comparative layer-wise analysis of self-supervised speech models," in *2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5.

[6] Minxue Niu, Najmeh Sadoughi, Abhishek Yanamandra, Pichao Wang, Zhu Liu, Vimal Bhat, and S. Elizabeth Norred, "Learning rich speech representations with acoustic-semantic factorization," in *Proc. IEEE ICASSP*, 2025.

[7] Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi, "Editing models with task arithmetic," in *International Conference on Learning Representations*, 2023.

[8] Zilai Wang, Natarajan Balaji Shankar, Kaiyuan Zhang, Zihan Wang, and Abeer Alwan, "Mind the shift: Using delta SSL embeddings to enhance child ASR," arXiv:2601.20142, 2026, Accepted at ICASSP 2026.

[9] Sheng-Chieh Chiu, Chia-Hua Wu, Jih-Kang Hsieh, Yu Tsao, and Hsin-Min Wang, "Learnable layer selection and model fusion for speech self-supervised learning models," in *Proc. Interspeech*, 2024, pp. 3914–3918.

[10] Pavlo Molchanov, Stephen Tyree, Tero Karras, Timo Aila, and Jan Kautz, "Pruning convolutional neural networks for resource efficient inference," in *International Conference on Learning Representations*, 2017.

[11] Alex Graves, Santiago Fernández, Faustino Gomez, and Jürgen Schmidhuber, "Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks," in *Proc. ICML*, 2006, pp. 369–376.

[12] Guanlong Zhao, Sinem Sonsaat, Alif Silpachai, Ivana Lucic, Evgeny Chukharev-Hudilainen, John Levis, and Ricardo Gutierrez-Osuna, "L2-ARCTIC: A non-native english speech corpus," in *Proc. Interspeech*, 2018, pp. 2783–2787.